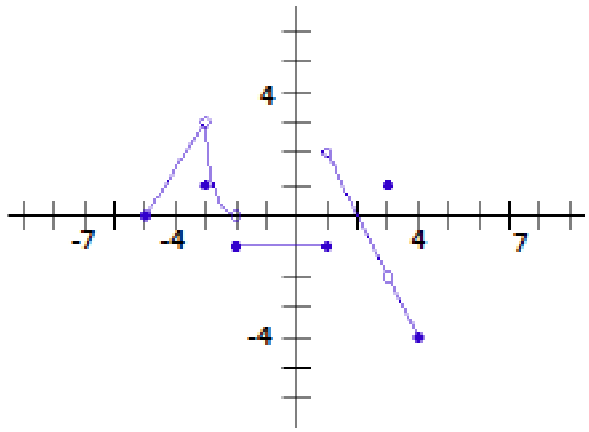
Your Name:

ID #:

Worksheet: Limits at Continuity (Discontinuity)-Part1

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Refer to the graph below, state with explanations whether the following limits exists.



Determine if the following limits exists:

$\lim\limits_{x\to-3} f(x)$ $\quad$ $\lim\limits_{x\to-2} f(x)$ $\quad$ $\lim\limits_{x\to0} f(x)$ $\quad$ $\lim\limits_{x\to1} f(x)$ $\quad$ $\lim\limits_{x\to2} f(x)$ $\quad$ $\lim\limits_{x\to3} f(x)$

2. (a) $\lim\limits_{x\to c}(2x+5) =$

   (b) $\lim\limits_{t\to 6} 8(t-5)(t-7) =$

   (c) $\lim\limits_{x\to 2} \dfrac{x+2}{x^2-5x+6} =$

   (d) $\lim\limits_{x\to 0} \dfrac{x^3+(x+1)ln(x+3)}{x^3-4} =$

   (e) $\lim\limits_{x\to 4} x^{x+1} =$

   (f) $\lim\limits_{x\to -2} e^{x+1} =$

   (g) $\lim\limits_{x\to \frac{\pi}{4}} sin(2x) =$

   (h) $\lim\limits_{x\to \frac{\pi}{2}} csc(x) =$

   (i) $\lim\limits_{x\to 0} e^{x} =$

   (j) $\lim\limits_{x\to -5} \dfrac{x^2+3x-5}{x+7} =$

   (k) $\lim\limits_{x\to 2} \dfrac{e^x}{x+1} =$

   (l) $\lim\limits_{x\to 3} \dfrac{x^2-3}{\ln e^x} =$

   (m) $\lim\limits_{x\to 0} \sqrt[3]{x^4+2x^3+8} =$

   (n) $\lim\limits_{x\to 0} (x^2+1)^{2x-2} =$

   (o) $\lim\limits_{x\to -1} e^{x^2} =$

   (p) $\lim\limits_{x\to \frac{\pi}{6}} \dfrac{tan(x)}{cos(2x)} =$

3. Show all the reasons to find the following limits:

   (a) $\lim_{x\to 0+} e^{\frac{1}{x}} =$

   (b) $\lim_{x\to 0-} e^{\frac{1}{x}} =$

   (c) $\lim_{x\to 0^+} lnx =$

   (d) $\lim_{x\to 0^-} xe^{\frac{1}{x}} =$

4. Show all the work to find the following limits:

   (a) $\lim_{x\to 0} \dfrac{\sin(3x)}{4x}$

   (b) $\lim_{x\to 1} \dfrac{5x^4-4x^2-1}{10-x-9x^3}$

   (c) $\lim_{x\to -1} \dfrac{\sqrt{x+4}-3}{x+1}$